

Larger contact recovery: retained full-trajectory results

The 267-row later hull system passes native and independent contact/passivity checks with residual **1.97e-12 m/s** after raising the numerical recovery row limit from 256 to 384. Its global recovery work budget remains 256 SVD calls; the successful captured-system replay needs four. This establishes a numerical capacity fix.

The separately frozen rotating/shaking 27-hull full-trajectory follow-up retains **three rejected attempts and no completed histories**. The later contact failures remain unqualified under unchanged accuracy and physical gates.

Attempt	Failed residual	Outcome
fast_rotate_shake27_hulls7301/reference_0	0.00254761 m/s	Rejected
fast_rotate_shake27_hulls7301/reference_1	0.00453103 m/s	Rejected
fast_rotate_shake27_hulls7301/reference_2	1.79796e-05 m/s	Rejected

Execution source: 4910fbca7cacf8dfd218be97a082d32b7ea13ea2. Authored geometry, density/mass/inertia, friction, restitution, gravity, prescribed 20 m/s translation/reversals and 10 rad/s rotation, timesteps and travel fractions are unchanged. Both earlier failed archives remain intact. The new capacity permits resolving one later system; subsequent unresolved systems still reject.

Sources, exact plan, authored scenes and every error are retained in **results/**. Audit with `python -m research.audit_spatial_recovery384`. The appended 267-row fixture and capture provenance are in **research/coulomb-diagnostics/**; its native replay is mandatory in CI. The 45-row second seed42 system remains unresolved. No claim of general random-hull feasibility, material authenticity, convergence order or arbitrary-body trajectory accuracy follows from these captured solves.